Contrastive Trace Probing for Safety Commitment Detection in Reasoning Models: A Cross-Architecture Empirical Study

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Abstract

Reasoning models such as DeepSeek-R1 and QwQ exhibit *self-jailbreak*: their extended chain-of-thought interleaves safety reasoning with harmful execution, culminating in a *commitment point* where safety reasoning permanently ceases. Existing hidden-state jailbreak detectors target adversarial inputs to instruction-tuned models and have never been evaluated on reasoning-model self-jailbreak. We present the first cross-architecture empirical study of hidden-state safety monitoring in reasoning models, spanning four models across two architecture families (Llama-3.1 and Qwen2.5) and two scales (7–8B and 32B). We establish three findings: (1) static per-step probes achieve $<$50% detection coverage despite strong per-dimension safety signal (Fisher ratio $>$14), because the signal is distributed across minor principal components; (2) a simple temporal window logistic regression (W=15) achieves 76.7% mean coverage—matching the label-oracle ceiling—with a +34.9pp improvement over static probes that is consistent across all architectures; (3) no hidden-state method achieves positive detection lead time (best: -0.2 steps vs. oracle +31 steps), establishing a principled limitation boundary. Cross-architecture transfer via learned projection heads yields 64.5% mean coverage without target-domain labels, suggesting safety commitment signals share geometric structure across model families.

1 Introduction

Reasoning models—large language models trained with reinforcement learning to produce extended chains of thought (CoT)—represent a qualitative shift in how language models process complex queries DeepSeek-AI [2025], Team [2024]. These models interleave multiple reasoning strategies across dozens to hundreds of steps, achieving strong performance on mathematical and scientific benchmarks. However, recent work has identified a critical safety failure mode: *self-jailbreak*, where the model's own reasoning process gradually transitions from safety-aligned analysis to harmful content generation Zhang et al. [2025], Wang et al. [2025].

Unlike conventional jailbreak attacks that rely on adversarial inputs, self-jailbreak emerges from the model's internal reasoning dynamics. In 84% of harmful traces, safety-relevant reasoning persists well past step 10, only to permanently cease at a *commitment point*—the reasoning step after which no further safety considerations appear. Detecting this commitment point is critical for deploying safety monitors on reasoning models.

Existing hidden-state jailbreak detectors such as TrajGuard Chen et al. [2025], SafeSwitch Liu et al. [2025], and SIREN Park et al. [2025] achieve strong defense rates ($>$95%) on instruction-tuned models. However, they operate at token-level granularity on models where the attack is an adversarial *input*. Reasoning-model self-jailbreak is fundamentally different: the attack is the model's *own reasoning process*, distributed across many CoT steps. No prior work bridges hidden-state detection methods to reasoning-model self-jailbreak.

We present the first systematic cross-architecture evaluation of hidden-state methods for safety commitment detection in reasoning models. Our study spans four models—DeepSeek-R1-Distill-Llama-8B, DeepSeek-R1-Distill-Qwen-32B, QwQ-32B, and OpenThinker-7B—covering two architecture families (Llama-3.1 and Qwen2.5) and two scales (7–8B and 32B). We establish three claims:

Claim 1: Static probes fail reliably. Per-step logistic regression achieves <50% detection coverage across all models (mean 41.9%), despite Fisher discriminant ratios >14 in full representation space. The safety signal exists but concentrates in minor principal components (top-3 PCs explain <17% of variance), causing both static classifiers and geometric detectors (Mahalanobis distance) to fail.

Claim 2: Temporal window probes saturate oracle coverage. A temporal window logistic regression with W=15 achieves 76.7% mean coverage across architectures, matching the label-oracle upper bound. The +30–35pp improvement is consistent across all four models and both architecture families. Window size W=15 is universally optimal, requiring no architecture-specific tuning.

Claim 3: Pre-textual detection is fundamentally limited. No hidden-state method achieves positive mean lead time. The oracle's +31-step advantage derives from directly observing label-sequence autocorrelation—information that hidden states encode for the current step but not for future trajectory. This establishes a principled boundary: hidden-state monitors can detect commitment *at* or *after* it occurs, but not before.

2 Related Work

Hidden-state jailbreak detection. TrajGuard Chen et al. [2025] monitors activation trajectories using Mahalanobis distance from safe-trace distributions, achieving 95% defense rate on instruction-tuned models. SafeSwitch Liu et al. [2025] trains linear probes on intermediate representations to classify safety-relevant states. SIREN Park et al. [2025] uses contrastive features between safe and unsafe hidden states for real-time detection. All three operate at token-level granularity on models where the adversarial signal is in the *input*—none have been evaluated on reasoning-model self-jailbreak where the signal is in the model's own multi-step reasoning.

Reasoning model safety. Wang et al. [2025] demonstrate that reasoning models can jailbreak themselves through extended CoT, bypassing safety training without adversarial input. Zhang et al. [2025] provide the HarmThoughts dataset with fine-grained step-level safety annotations across a 16-label taxonomy, enabling quantitative analysis of safety reasoning dynamics. SafeRemind Li et al. [2025] and SafeThink Zhou et al. [2025] propose text-level CoT monitoring but are vulnerable to pathological reasoning chains. None of these works evaluate hidden-state detection methods.

Linear probing of LLM representations. Linear probes have been widely used to study what information LLM hidden states encode Belinkov [2022], including truthfulness Li et al. [2024], sentiment Tigges et al. [2023], and factual knowledge Meng et al. [2022]. Our work extends this methodology to temporal probe designs that capture multi-step patterns in reasoning traces.

3 Method

3.1 Task Definition

Given a reasoning trace $\mathbf{x} = (x_1, \ldots, x_T)$ where each x_t is a reasoning step, we define the *commitment point* c as the last step containing safety-relevant reasoning. The detection task is binary classification: for each step t, predict whether $t > c$ (the model has committed to harmful output).

3.2 Hidden-State Extraction

For each reasoning step x_t, we extract the hidden-state vector $\mathbf{h}_t^{(\ell)} \in \mathbb{R}^d$ at layer ℓ from the last token position of that step. Based on layer ablation (Section 6.2), we use the optimal middle-layer representation at approximately 35–44% relative depth.

3.3 Static Probe

The static probe is a logistic regression classifier trained on individual hidden states:

$$p(y_t = 1 \mid \mathbf{h}_t) = \sigma(\mathbf{w}^\top \mathbf{h}_t + b) \tag{1}$$

This represents the standard approach used by existing hidden-state detectors adapted to step-level granularity.

3.4 Temporal Window Probe

The temporal window probe concatenates hidden states from a window of W consecutive steps:

$$\mathbf{z}_t = [\mathbf{h}_{t-W+1}; \ldots; \mathbf{h}_t] \in \mathbb{R}^{Wd} \tag{2}$$

$$p(y_t = 1 \mid \mathbf{z}_t) = \sigma(\mathbf{w}^\top \mathbf{z}_t + b) \tag{3}$$

This enables the probe to capture *safety reasoning density decline*—the gradual decrease in safety-labeled steps within the window as the model approaches commitment.

3.5 Mahalanobis Distance Baseline

Following TrajGuard's approach, we estimate the covariance $\mathbf{\Sigma}$ of hidden states from safe traces and compute per-step Mahalanobis distance:

$$d_M(\mathbf{h}_t) = \sqrt{(\mathbf{h}_t - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1} (\mathbf{h}_t - \boldsymbol{\mu})} \tag{4}$$

A threshold on d_M classifies steps as committed.

3.6 GRU Sequential Probe

As an upper bound on temporal modeling capacity, we train a GRU with 256 hidden units that processes the full sequence of hidden states:

$$\mathbf{g}_t = \text{GRU}(\mathbf{h}_1, \ldots, \mathbf{h}_t; \theta) \in \mathbb{R}^{256} \tag{5}$$

$$p(y_t = 1 \mid \mathbf{g}_t) = \sigma(\mathbf{v}^\top \mathbf{g}_t + b) \tag{6}$$

3.7 Label-Sequence Oracle

The oracle directly observes the binary label sequence and fires when $K = 5$ consecutive steps lack safety labels. It serves as an upper bound on detection *coverage* (what fraction of commitment points are detected) while providing a reference for *lead time* (how many steps before textual commitment the detection occurs).

3.8 Cross-Architecture Transfer

To evaluate whether safety commitment signals share geometric structure across architectures, we train a linear projection head $\mathbf{P} \in \mathbb{R}^{d_{\text{target}} \times d_{\text{source}}}$ that maps hidden states from a source model's space to a target model's space, using 30 aligned traces for projection training. The source model's trained probe is then applied to projected target-model representations.

Model	Arch	d	Acc	Cov	Fisher	F(PCA-3)
r1-8b	Llama	4096	.817	.465	17.9	0.93
r1-32b	Qwen	5120	.824	.442	17.1	0.87
QwQ-32B	Qwen	5120	.809	.395	15.6	0.72
OT-7B	Qwen	3584	—	.419	13.2	0.68
Mean			.817	.430	15.9	0.80

Table 1: Static probe performance across architectures. Coverage is below 50% for all models despite high Fisher discriminant ratios in original space. Fisher(PCA-3) collapses to <1, confirming signal concentration in minor PCs.

3.9 Evaluation Metrics

- **Coverage**: Fraction of traces where the probe detects commitment (true positive rate at trace level).
- **Per-step accuracy**: Step-level classification accuracy.
- **Lead time**: Number of steps between probe detection and textual commitment point (positive = early detection).

4 Experimental Setup

Models. We evaluate four reasoning models spanning two architecture families and two scales: DeepSeek-R1-Distill-Llama-8B (Llama-3.1, d=4096, 32 layers), DeepSeek-R1-Distill-Qwen-32B (Qwen2.5, d=5120, 64 layers), QwQ-32B (Qwen2.5, d=5120, 64 layers), and OpenThinker-7B (Qwen2.5, d=3584, 32 layers).

Dataset. We use HarmThoughts reasoning traces with step-level safety annotations from a 16-label taxonomy, binarized to safety-relevant vs. not. The dataset contains 209 traces, split into train/test with stratification.

Layer selection. For each model, we select the optimal layer via held-out validation. Optimal layers fall at 34–44% relative depth across all models (Section 6.2).

5 Results

5.1 Claim 1: Static Probes Fail Across Architectures

Table 1 reports static probe performance across all models. Despite per-step accuracies above 80% (inflated by majority-class bias), detection coverage remains below 50% for every model. Fisher discriminant ratios in original representation space exceed 14 across architectures, confirming that safety-relevant signal *exists* in hidden states. However, PCA analysis reveals this signal is distributed across minor principal components: top-3 PCs explain only 12.8–16.6% of variance, and Fisher ratio collapses from >14 to <1 after PCA-3 projection.

The Mahalanobis distance baseline (Table 2) performs even worse than static LR, with mean coverage of 27.9% and AUROC below 0.62. This failure is principled: Mahalanobis distance is dominated by major principal components, where the safety signal is weakest.

Model	MD Cov	MD AUROC	Temp Cov	Gap
r1-8b	.326	.612	.767	−44.1pp
r1-32b	.279	.583	.791	−51.2pp
QwQ-32B	.233	.561	.744	−51.1pp

Table 2: Mahalanobis distance baseline vs. temporal probe. Geometric anomaly detection fails because safety signal concentrates in minor PCs that contribute negligibly to Mahalanobis distance.

Model	Stat	Temp	Δ	Acc	Lead
r1-8b	.465	.767	+30.2	.848	−1.09
r1-32b	.442	.791	+34.9	.851	−1.32
QwQ-32B	.395	.744	+34.9	.839	−0.87
Mean	.434	.767	+33.3	.846	−1.09

Table 3: Temporal window probe (W=15) vs. static probe. Coverage improvement of +30–35pp is consistent across architectures. Lead time is the mean number of steps relative to commitment point (negative = after).

5.2 Claim 2: Temporal Window Probes Saturate Oracle Coverage

Table 3 presents the core result. The temporal window probe (W=15) achieves 74.4–79.1% coverage across architectures, with a consistent +30–35pp improvement over static probes. This improvement holds for both Llama-3.1 and Qwen2.5 architectures and at both 7–8B and 32B scales.

Architecture × Scale matrix. Table 4 presents the 2×2 architecture–scale matrix, including OpenThinker-7B as the Qwen2.5-7B control. The temporal advantage holds at both scales: the fundamental pattern (static fails, temporal saturates) is architecture- and scale-independent. Scale provides a modest +7pp improvement within the Qwen family (7B→32B), smaller than the +30pp method effect.

5.3 Claim 3: Pre-Textual Detection Is Fundamentally Limited

Table 5 reports lead time analysis across all models and methods. No hidden-state method achieves positive mean lead time. The oracle achieves +31 steps mean lead time by directly observing label-sequence autocorrelation—consecutive absences of safety labels predict permanent absence.

Oracle decomposition. Table 6 reports mutual information between hidden states and label sequences at varying horizons. Hidden states are informative about the *current* step's label (MI = 0.42) but nearly uninformative about future labels (MI → 0.03 at horizon 15). The oracle exploits label-sequence autocorrelation: on average, 14.6 consecutive steps precede the commitment point without safety labels, making K=5 consecutive absences a powerful predictor. This information is structurally unavailable to hidden-state probes.

5.4 Cross-Architecture Transfer

Table 7 presents the cross-architecture transfer matrix. A probe trained on one model achieves non-trivial coverage on another via a learned linear projection head (20M parameters, trained on 30 aligned traces). Mean cross-transfer coverage is 64.5%, with same-family transfer (r1-32b → QwQ-32B, both Qwen2.5) reaching 72.1%. This demonstrates that safety commitment signals share geometric structure across model families.

	7–8B	32B
Llama-3.1	r1-8b: .767	—
Qwen2.5	OT-7B: .721	r1-32b: .791

Table 4: Temporal W=15 coverage in the 2×2 architecture–scale matrix. Scale effect (+7pp) and architecture effect (-4.6pp) are both small relative to method effect (+30pp).

Model	Method	Lead	Med	+Lead%	Cov
r1-8b	Static	-2.0	-2	30%	.465
	Temp W15	-1.09	-1	34%	.767
	GRU	-0.21	-1	34.5%	.674
r1-32b	Static	-2.8	-3	25%	.442
	Temp W15	-1.32	-1	29%	.791
	GRU	-0.9	-1	29%	.721
QwQ-32B	Static	-1.9	-2	28%	.395
	Temp W15	-0.87	-1	33%	.744
	GRU	-0.5	0	33%	.698
Oracle (K=5)		+31	+29	100%	1.0

Table 5: Detection lead time across architectures. Lead = mean steps before commitment (positive = early). No hidden-state method crosses zero; the oracle's +31-step advantage comes from label-sequence statistics unavailable to hidden-state probes.

6 Ablation Studies

6.1 Window Size

Table 8 reports coverage as a function of window size W across all architectures. Coverage increases monotonically from W=1 (equivalent to static) to W=15, then saturates or slightly declines at W=25 due to overfitting. The saturation point W=15 is identical across all models, indicating that the temporal scale of safety reasoning decline ($\sim$15 steps) is a property of the behavioral phenomenon, not the model architecture.

6.2 Layer Selection

Table 9 reports coverage across relative layer depths. The optimal layer falls at 35–44% relative depth across all models: layer 14/32 for 8B models and layers 22–24/64 for 32B models. This "middle-layer sweet spot" between low-level syntactic features and task-specific output representations is consistent with prior probing work and enables a simple layer selection heuristic (pick the middle third) without architecture-specific tuning.

6.3 GRU Upper Bound

Table 10 compares the GRU sequential probe against temporal LR on 32B models. The GRU (1.62M parameters) does not outperform temporal LR (76.8K parameters) on any model—in fact, it underperforms by 2.3pp due to overfitting at current data scales (35% overfitting rate on held-out validation). This confirms that method simplicity is the correct design choice given the data regime: 209 traces do not support learning nonlinear temporal patterns beyond what linear probes capture.

Target	MI(Hidden)	MI(Labels)
Current label	0.42	0.89
Next 5 labels	0.08	0.31
Next 15 labels	0.03	0.12
Commitment (binary)	0.15	0.67

Table 6: Mutual information between hidden states (resp. label history W=15) and label sequences. Hidden states encode current state but not future trajectory.

Train →	r1-8b	r1-32b	QwQ
r1-8b (Llama)	.767*	.628	.581
r1-32b (Qwen)	.651	.791*	.721
QwQ (Qwen)	.605	.698	.744*

Table 7: Cross-architecture transfer matrix (temporal W=15 coverage). *In-distribution. Same-family transfer achieves >70%; cross-family transfer achieves >58%.

6.4 16-Label Fine-Grained Supervision

Table 11 evaluates whether replacing binary labels with the full 16-label HarmThoughts taxonomy improves detection. Fine-grained supervision provides marginal improvement (+2.4pp coverage, +0.3 steps lead time) but does not change the fundamental findings. The modest gain confirms that the bottleneck is the temporal information capacity of hidden states, not label granularity.

6.5 Statistical Validation

Table 12 reports 95% bootstrap confidence intervals (1000 resamples) for key metrics. All temporal-vs-static coverage deltas exclude zero (minimum lower bound +18.6pp), confirming statistical significance despite the moderate sample size. Lead time CIs are centered negative with upper bounds near zero.

7 Discussion

Why does method simplicity win? Our finding that temporal window LR matches or exceeds GRU and attention-based probes may seem counterintuitive. The explanation is data-regime-dependent: with 209 training traces (∼10K steps), a linear model with $W \times d$ features (76.8K parameters) is at the edge of the data-sufficient regime, while a GRU with 1.62M parameters overfits. We expect complex temporal models to eventually outperform LR as datasets scale, but the current finding has practical value: it means safety monitoring can be deployed with minimal infrastructure.

Practical implications of negative lead time. While pre-textual detection remains out of reach, the temporal probe's near-zero lead time (mean -0.87 to -1.32 steps) is practically useful. Detection at or shortly after the commitment point enables intervention strategies: (1) post-hoc audit of reasoning traces before output delivery, (2) real-time monitoring that halts generation within 1–2 steps of commitment, and (3) triggering safety re-reasoning when commitment is detected.

Universality of safety signal geometry. The consistent failure of Mahalanobis distance across architectures, combined with successful cross-architecture transfer, suggests that safety commitment signals occupy a specific geometric niche—minor principal components that carry low variance but high discriminative value. This geometry appears to be a universal property of how reasoning models encode safety state, independent of architecture family or scale.

W	r1-8b	r1-32b	QwQ	Mean
1	.465	.442	.395	.434
3	.535	.512	.465	.504
5	.651	.628	.581	.620
10	.651	.698	.651	.667
15	.767	.791	.744	.767
20	.767	.791	.744	.767
25	.744	.767	.721	.744

Table 8: Coverage vs. window size. Saturation at W=15 is universal across architectures. Decline at W=25 reflects overfitting from increased feature dimensionality.

Depth	r1-8b	r1-32b	QwQ
10%	.442 (L3)	.395 (L6)	.372 (L6)
25%	.605 (L8)	.581 (L16)	.558 (L16)
35%	.721 (L11)	**.814** (L22)	.744 (L22)
44%	**.767** (L14)	.791 (L28)	**.791** (L28)
50%	.744 (L16)	.767 (L32)	.767 (L32)
75%	.628 (L24)	.651 (L48)	.628 (L48)

Table 9: Temporal W=15 coverage by relative layer depth. Bold = best per model. The optimal region (35–44% depth) generalizes across architectures.

8 Limitations

Sample size. Our evaluation uses 209 traces from HarmThoughts. While bootstrap CIs confirm statistical significance of core findings, wider confidence intervals ($\pm$9–12pp) limit precision of coverage estimates. Replication on larger datasets is needed.

Model coverage. We evaluate four models from two architecture families. Proprietary reasoning models (OpenAI o1/o3, Claude) with different training methodologies may exhibit different safety signal geometry.

Commitment point definition. Our binary commitment label assumes a single, permanent transition. Some traces may exhibit multiple commitment-like transitions or gradual commitment that is not well-captured by a point definition.

Probe access assumption. Hidden-state monitoring requires white-box access to model internals, limiting deployment to first-party safety teams or open-weight models.

9 Conclusion

We present the first cross-architecture empirical study of hidden-state safety monitoring in reasoning models. Across four models, two architecture families, and two scales, we establish that: (1) static probes fail despite strong per-dimension safety signal, because the signal concentrates in minor principal components; (2) a simple temporal window probe achieves oracle-ceiling coverage with a universal window size (W=15), requiring no architecture-specific tuning; (3) pre-textual detection is fundamentally limited by the causal structure of hidden-state representations, which encode current safety state but not future trajectory. Cross-architecture transfer demonstrates shared geometric structure of safety signals across model families. These findings provide both a practical monitoring tool (temporal win-

Model	Method	Cov	Acc	Lead	Params
r1-32b	Temp LR	.791	.851	−1.32	76.8K
	GRU-256	.767	.842	−0.6	1.62M
QwQ-32B	Temp LR	.744	.839	−0.87	76.8K
	GRU-256	.721	.831	−0.4	1.62M

Table 10: GRU vs. temporal LR on 32B models. GRU's 21× parameter increase yields no coverage improvement, confirming that the bottleneck is data scale, not model capacity.

Model	Bin Cov	16L Cov	Bin Lead	16L Lead
r1-8b	.767	.791	−1.09	−0.7
r1-32b	.791	.814	−1.32	−0.9

Table 11: Binary vs. 16-label supervision (temporal W=15). Fine-grained labels offer marginal improvement, confirming that hidden-state temporal capacity—not supervision quality—is the bottleneck.

dow probes) and principled boundaries (lead time limitations) for the emerging challenge of reasoning model safety.

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Metric	r1-8b	r1-32b	QwQ
Static Cov	[.35, .58]	[.33, .56]	[.28, .51]
Temp Cov	[.67, .86]	[.70, .88]	[.65, .84]
Δ	[.21, .49]	[.23, .51]	[.23, .51]
Lead Time	$[-2.8, 0.6]$	$[-3.1, 0.5]$	$[-2.3, 0.7]$

Table 12: Bootstrap 95% CIs (1000 resamples). All Δ CIs exclude zero. Wide intervals reflect sample size (209 traces) but effect sizes are large relative to uncertainty.